

Sweta Ashara

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SKILLS

Soft Skills: Analytical Thinking, Problem-Solving, Communication, Leadership, Decision-Making, Collaboration, Strategic Planning

Quality & Regulatory Standards: ISO 13485, ISO 9001, 21 CFR 820, IATF 16949, GMP, GLP, ASTM Standards, Design Control, Verification & Validation (V&V), PPAP, FMEA, Gage R&R, GD&T Basics, Engineering Drawings, Risk Management, Root Cause Analysis (RCA), SPC

Data Analytics & Visualization: Power BI, Tableau, Power Apps, Power Automate, SQL, Python (Pandas, NumPy), DAX, Minitab, Excel (Pivot Tables, Macros, VBA), Data Modeling, Data Visualization, Statistical Analysis, Data Engineering (ETL, cleaning, transformation, integration), Dashboard Development, KPI Tracking, Real-Time Monitoring

Quality Tools & Methods: SCARs, CAPA, Supplier Audits, Supplier Qualification/Requalification, Approved Supplier List (ASL) Management, Vendor Performance Metrics, Supplier Scorecards, Non-Conformance Management, Cross-Functional Collaboration, Continuous Improvement, Risk Mitigation

Project & Process Management Tools: Jira, Confluence, Agile Project Management, SAP

EXPERIENCE

Quality and Process Control Engineer, CertainTeed LLC, Hagerstown, MD

June 2024-Oct 2025

- Developed and implemented a defect containment and NCR management system with automated investigation triggers, integrating CAPA and RCA workflows to ensure timely root cause identification, corrective/preventive actions, and improved product quality.
- Developed Python/SQL scripts to extract, clean, and analyze data from PLC Tags and SCADA systems, integrating the results into Power BI dashboards (COPQ, Raw Material COA, MES, PLC/SCADA Data) to enhance data accessibility, automation, and real-time, data-driven decision-making.
- Led the planning and implementation of sensor installations across the production line, managing supplier sourcing, budgeting, and technical evaluations to monitor CTQ (Critical to Quality) parameters and enhance process control.
- Created comprehensive Process Flow Diagrams (PFDs) and conducted Failure Mode and Effects Analysis (FMEA) to proactively detect and address potential process failures and ensure robust process reliability.
- Designed and implemented validation protocols and Site Acceptance Testing (SAT) to verify system functionality, ensure regulatory compliance, and maintain process reliability.
- Designed and deployed a PowerApps application integrated with existing data systems to monitor hazardous chemical inventory, manage waste disposal workflows, and maintain EHS compliance.
- Drove continuous improvement projects using Statistical Process Control (SPC) analysis, to reduce material variance, improve thickness control, and deliver significant cost savings and improve overall production/ manufacturing efficiency (OEE).
- Managed document control and quality assurance activities, including creating and updating procedures and work instructions, supporting internal and ISO 9001 audits, and assisting in management reviews to ensure compliance and continuous improvement.
- Supported the rollout of Industry 4.0 manufacturing systems and Canary by collecting shop-floor feedback, submitting improvement tickets, and designing real-time process dashboards to enhance visibility, monitoring, and data-driven decision-making.
- Managed supplier quality for multiple projects, including vendor selection, technical evaluations, and ongoing performance monitoring to ensure compliance with specifications and timely delivery of critical components
- Supported cross-functional projects by providing supplier data analysis, quality metrics, and process recommendations, enabling data-driven decision-making and risk mitigation

QA Engineer, Tesla Inc, Austin, TX

Nov 2023-June 2024

- Executed internal audits and CAPA assessments in compliance with ISO 9001 and IATF 16949, ensuring adherence to QMS requirements and 8D problem-solving methodology.
- Developed and maintained Process Flow Diagrams (PFDs) for automotive manufacturing lines, defining inline vs offline inspections and supporting defect prevention strategies.
- Designed, developed and implemented a software-based containment system to issue NCRs, eliminate nonconforming product from reaching customers, and minimize field failures and recalls.
- Conducted Root Cause Analysis (RCA) for process and product non-conformances, implementing corrective actions to prevent recurrence.
- Updated and standardized QMS documentation and conducted training sessions to ensure team compliance with automotive quality procedures.
- Collaborated closely with internal customers and suppliers to ensure quality product delivery, resolve issues, and prevent recurrence
- Extracted, cleaned, and analyzed large-scale manufacturing datasets using SQL to uncover trends, identify high-impact issues, and support operational decision-making

Quality Manufacturing Engineer, Zeda Health Inc, Cincinnati, OH

Sept 2022-Nov 2023

- Designed and improved, and standardized manufacturing processes using lean methodologies to optimize efficiency, reduce waste, and enhance product quality while adhering to customer requirements and industry standards (ISO 13485 and 21 CFR 820).
- Developed and maintained comprehensive process documentation including manufacturing process flows, bills of materials (BOM), work instructions, control plans, and inspection reports
- Analyzed production data and utilized quality tools such as process capability analysis, Failure Mode and Effect Analysis (FMEA), and statistical process control (SPC) to analyze and improve process performance, minimize defects, and boost customer satisfaction.
- Developed and implemented inspection and testing plans for in-process and finished goods, including First Article Inspection (FAI) forms, and automated data population using Macros/VBA.
- Collaborated closely with customers, vendors, and internal cross-functional teams, ensuring manufacturability and customer satisfaction.
- Developed validation protocols (IQ/OQ/PQ) and validation procedures for Additive Manufacturing and CNC machining Equipment.
- Conducted root cause analysis, corrective and preventive action (CAPA) investigations to identify and eliminate sources of nonconformance (NCR).
- Collaborated closely with customers during the prototyping phase through design transfer, ensuring manufacturability, regulatory compliance, and customer satisfaction.
- Conducted on-site supplier audits, assessing manufacturing processes, quality systems, and regulatory compliance, and implemented corrective actions for non-conformances.
- Worked extensively with external vendors for post-processing of manufactured devices, ensuring quality, consistency, and adherence to ASTM or customer-specific standards.
- Conducted meticulous Gage R&R studies to assess measurement system variability and ensure precision in manufacturing processes.
- Conducted and led internal and external audits to ensure compliance with regulatory standards and customer requirements, prepared audit/summary reports, and collaborated with the team on mitigation actions.

Quality Assurance Engineer, Lantz Medical, Zionsville, IN

Aug 2020- Sept 2022

- Remediated 2 USFDA Warning Letters by reverse engineering the product and creating retrospective DHF, DMR, and DHR documentation. Performed design and software validation and verification (V&V), authorizing precise testing protocols and test methods
- Optimized and implemented quality procedures, conducted, and supported internal audits, created automated workflows, and improved various processes in the manufacturing facility.
- Oversaw supplier quality programs, including evaluations, audits, maintaining Approved Supplier Lists (ASL), vendor agreements, and performance monitoring.
- Investigated customer complaints and implemented an effective CAPA and non-conformance identification system.
- Served as a project/quality manager in new product development, ensuring compliance with design controls requirements by identifying project requirements, developing plans, managing, and communicating with vendors, allocating resources, reviewing milestones, creating, and executing product validation plans and design control documentation for successful product launch.
- Developed multiple software application systems for company processes such as complaints, maintenance, shipping, and receiving, revolutionizing data collection, and enabling data modeling, analysis, and visualization, to further open any corrective or preventive action to improve processes, reduce waste and increase efficiency.
- Designed queries and dashboards for management review, providing valuable insights, relationships, and growth recommendations.
- Collaborated seamlessly with diverse departments to optimize workflows and enhance internal system integration.
- Designed and deployed multiple applications system software for warehouse inventory management, including material inventory.

EDUCATION

Purdue University, Indianapolis

May 2020

Master of Science (MS), Biomedical Engineering, 3.4 GPA

Mumbai University, Mumbai, India

May 2017

Bachelor of Science (BS), Biomedical Engineering, 3.7 GPA

Lean Six Sigma Black Belt

July 2022

ISO 13485 Lead Auditor Training

Nov 2022